Stuti Handa

+91 6239847492 | stutihanda6@gmail.com | LinkedIn | GitHub

Education

Vellore Institute of Technology Chennai, India
B.Tech in Computer Science Engineering (Data Science) ; CGPA: 8.3 *2023 – 2027*

Projects

Climate Guardian AI GitHub
Python, FastAPI, Streamlit, Plotly, XGBoost, LightGBM, SHAP, n8n, Ollama, SQLite, Open-Meteo API

- Built a full-stack, agent-driven climate risk intelligence platform that ingests environmental telemetry (temperature, humidity, precipitation) and predicts regional outbreak/hazard risk levels.
- Engineered an AutoML pipeline orchestrator that trains and benchmarks five models (Logistic Regression, Decision Tree, Random Forest, XGBoost, LightGBM) and auto-selects the best performer.
- Designed an interactive Streamlit command-center with KPI dashboards, correlation heatmaps, 3D cluster visualization, and SHAP-based feature explainability charts.
- Integrated a conversational AI Risk Assistant via an **n8n workflow calling a local Ollama LLM**, with live weather lookups (Open-Meteo API) and a bcrypt-authenticated SQLite user store.

Personal AI Assistant GitHub
n8n, Ollama, Groq, Streamlit, SerpAPI, Google Calendar, Gmail, Google Docs, Google Sheets, Google Tasks

- Built an end-to-end agentic personal assistant with a Streamlit chat UI and n8n webhook-triggered AI agent, supporting session-aware multi-turn conversations via window memory.
- Integrated 6 tool categories into a single ReAct agent: web search (SerpAPI), Gmail read/fetch, Google Calendar CRUD, Google Tasks (create/get/list/delete), Google Docs (create/read/update), and expense tracking via Google Sheets.
- Configured dual LLM backends — **Ollama** (local) and **Groq** (cloud) — with hot-swap capability; exposed via a REST webhook consumed by the Streamlit frontend.

AnalyzeProject – Research Paper Publishability Prediction GitHub
Python, XGBoost, scikit-learn, spaCy, NLTK, KeyBERT, PyMuPDF, pandas, NumPy

- Built an ML pipeline to predict research paper publishability using 16 engineered features — readability, structural indicators, and citation quality; achieved **94.4% test accuracy** with XGBoost.
- Identified Figures-to-Text Ratio (**21.8%**) and Title–Abstract Consistency (**11.4%**) as top predictors using PyMuPDF, spaCy, NLTK, and KeyBERT for feature extraction.

Position of Responsibility

Wakhra Punjab Club – VIT Chennai 2024 – 2025
Design Team *Chennai, India*

- Designed and executed artistic installations for **Vibrance 2025**, VIT's annual cultural fest with **30,000+ attendees**; led stage decor and branding in collaboration with cross-functional teams.

Technical Skills

Languages: C++, Python, SQL, R, Java, JavaScript, HTML, CSS

Machine Learning & DL: Scikit-learn, XGBoost, LightGBM, TensorFlow, Feature Engineering, NLP, SHAP

Data Science: Pandas, NumPy, Matplotlib, Seaborn, Plotly, Statistical Analysis, EDA

Generative AI & LLMs: LLMs, Transformers, RAG, Prompt Engineering, ReAct, LangChain, LangGraph, Vector DBs, Ollama

Backend & APIs: FastAPI, Pydantic, REST APIs, Webhooks, JSON, Streamlit

Automation & Integration: n8n, Workflow Automation, API-based Integrations

Databases & Cloud: SQLite, PostgreSQL, MySQL, AWS Fundamentals

Tools: Git, GitHub, Docker, Google Colab, VS Code, Jupyter Notebook

Certifications

- AI-102: Microsoft Certified Professional (MCP) – Azure AI Engineer Associate
- Legacy Responsive Web Design Certification – freeCodeCamp